Imran Ali

AI Engineer | Backend Developer

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Education

BSc Computer Science **FAST NUCES CFD Campus, Pakistan** *August 2022 – June 2026*

Relevant Courses: Deep Learning, NLP, Generative AI, Web Programming, Information Security, Software Engineering

Experience

Data Analyst Intern ***Brickclay*** *June 2025 – August 2025*

- Built an end-to-end ETL pipeline to ingest and transform structured data into SQL Server for downstream analytics.
- Architected a star-schema dimensional model (fact and dimension tables) for the Rental Open Dataset, reducing query complexity and improving reporting performance.
- Built interactive Power BI dashboards with KPIs and drill-down analysis, enabling data-driven business decisions across departments.

Skills

- **AI & Deep Learning:** PyTorch, HuggingFace, Fine-tuning, RAG, LangChain, vector databases, LLM APIs, Vision Transformer, Qwen2-VL, Gemini API
- **Languages & Databases:** Python, C++, OracleSQL, MS SQL Server, MongoDB
- **Tools & Frameworks:** Docker, Git, GitHub, Selenium, Playwright, FastAPI, RESTful APIs, CI/CD, pytest

Projects

DiReCT - Clinical RAG for Diagnostic Reasoning GitHub *(FastAPI, React, LangChain, FAISS, BM25, LLM APIs)*

- Built an end-to-end Clinical RAG system over 511 real MIMIC-IV physician notes with hybrid retrieval (BM25 + FAISS fused via RRF), cross-encoder reranking (ms-marco-MiniLM-L-6-v2), and query expansion to MIMIC-style vocabulary. Dual LLM generation (Gemini 2.5-flash + Groq fallback) returns structured responses with citations and critical flags; achieved Retrieval Precision@5: 0.64 and Generation score: 4.93/5.0.

Document to Markdown Generation (VLM Fine-tuning) Demo *(Qwen2-VL, QLoRA, PEFT)*

- Fine-tuned Qwen2-VL-2B-Instruct on 14,236 image-markdown pairs (Nougat Dataset) using QLoRA (4-bit NF4) on dual T4 GPUs for document understanding — headings, citations, multi-column layouts, and structured markdown generation; achieved BLEU: 0.84, ROUGE-L: 0.92, Train Loss: 0.07.

TestHub (AI-Powered Test Automation Platform) GitHub *(Go, Python, Docker, Playwright, Selenium, LLM APIs)*

- Built a scalable web-based test automation platform supporting parallel test execution across Chrome and Firefox using Selenium Grid and Playwright, with Python and Java support, artifact storage, and reporting.
- Implemented an AI-powered test script generator that scrapes web pages using Playwright, preprocesses DOM data, and generates executable test scripts via LLMs (Gemini with Groq fallback), integrating seamlessly into the existing execution pipeline with validation.

Neural Storyteller: Image Captioning Demo *(PyTorch, torchvision)*

- Built an end-to-end image captioning pipeline using ResNet50 for feature extraction and an LSTM-based Seq2Seq decoder. Implemented greedy decoding and beam search strategies with a Gradio web app for real-time inference.

Denoising Diffusion Probabilistic Model (DDPM) Demo *(PyTorch, U-Net, Gradio)*

- Built a DDPM from scratch with U-Net backbone ($64 \rightarrow 512$ channels): self-attention at 16×16 and 32×32 resolution, sinusoidal time embeddings, and residual skip connections. Trained on 30k FFHQ faces for 60 epochs using mixed precision on dual T4 GPUs achieving PSNR 11.36 dB and SSIM 0.21.

Pix2Pix GAN - Anime Sketch Colorization Demo *(PyTorch, U-Net, PatchGAN)*

- Trained a GAN (U-Net Generator + PatchGAN Discriminator) on 14k+ sketch-color pairs with Adversarial + L1 Loss on dual T4 GPUs. Achieved SSIM 0.64, PSNR 14.58 dB — deployed live on Hugging Face for real-time colorization.

Achievements & Certifications

Introduction to Docker — *DataCamp* | **The Git & GitHub Bootcamp** — *Udemy*

LeetCode: Solved 100+ problems in Data Structures & Algorithms

Hugging Face: Deployed 10+ end-to-end AI project demos on Hugging Face Spaces